Asteroids and other Celestial Bodies - Some Legal Differences

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I. Introduction

Since the beginning of the work on Space Law the term "Celestial Bodies" was used. This expression begins with the Czech author Vladimir Mandl in 1932, the very first writer dealing with Outer Space Law,[1] and is followed up by the writings in the 1950's and early sixties.[2] When the United Nations started their investigation of the related problems, the topic of Celestial Bodies was one of them, and not the least one.

Proposals of the USA of Nov. 13, 1957[3] and the (then) USSR of March 15, 1958[4] both proposed (in different terms) the use of Outer Space for peaceful - or nonmilitary use. Celestial Bodies were not mentioned specifically. Neither were such Bodies mentioned in the (not unanimous) UNGA Resolution of Dec. 13., 1958[5] which established the *ad hoc Committee on the Peaceful Uses of Outer Space,* and which requested the Peaceful Uses of Outer Space.

The UNGA Resolution of Dec. 12, 1959,[6] which created a (not *"ad hoc"*) Committee on the Peaceful Uses of Outer Space (COPUOS) requested the said Committee to "study the nature of legal problems which may arise from the exploration of Outer Space," but did not mention Celestial Bodies either.

There followed the unanimous UNGA Resolution of Dec. 20, 1961,[7] already laying down some principles of Space Law, notably that: *"Outer Space and Celestial Bodies are free for Exploration and Use by all States in conformity with international Law, and are not subject to national appropriation."*

Now it was obvious that Space Law should apply to Celestial Bodies, and that they were (obviously) not the same as the empty (or nearly empty) Outer Space itself. This notwithstanding the fact that the so called "Test Ban Treaty," signed in Moscow on August 5, 1963, still spoke about

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1 DAS WELTRAUMRECHT, EIN PROBLEM DER RAUMFAHRT, (Bensheimer Verlag, Berlin/Mannheim, 1932). A new edition is being prepared by Dr. Stephen Doyle.

2 *See*, for instance, LEGAL ASPECTS OF SPACE EXPLORATION, A SYMPOSIUM, (87th US Congress, 1st Sess., US Senate Doc. No. 26, Washington, D.C., 1961) and ERNST FASAN, WELTRAUMRECHT (Krausskopf Flugwelt Verlag, Mainz 1965) as well as the literature quoted therein.

3 U.N. Doc. A/C 12PV. 821,6.

4 U.N. Doc. A 3818.

5 U.N.G.A. Res. 1348(13).

6 U.N.G.A. Res. 1472 (14).

7 U.N.G.A. Res. 1721(16).

"nuclear weapons explosion ... in outer space ...", not mentioning celestial bodies specifically.[8]

The following UNGA Resolutions, however, continued to uphold the notion of Celestial Bodies in general,[9] although some scholars such as Cocca,[10] Haley[11] and the Hon. George Miller,[12] House of Representatives of California, pointed out that the Moon might have special legal characteristics.

But then, the "Treaty on Principles, Governing the Activities of States in Exploration of Outer Space, including the Moon and other Celestial Bodies" (The Space Treaty)[13] [14] stated in its Art. II: "Outer Space, including the Moon and other Celestial Bodies, are not subject to national appropriation by claim of sovereignty, by means of use or occupation or by any other means."

II. Astronomical Facts about Celestial Bodies.

We know, and we have to take into account, that there are several kinds of Celestial Bodies. Our Universe shows Billions of Galaxies, each of which consists of billions of stars. One of them is our own Galaxy, also called "The Milky Way." And it was made clear by the then President of the United States, Mr. Lyndon B. Johnson, that our Galaxy is the realm of the terrestrial nations exploration.[15] Stars are huge and enormously hot bodies, and some of them seem to be surrounded by planet-like subsidiaries. With the exception of the sun they are many light years away, so any question of appropriation of stars need not be considered at the present time and the foreseeable future. But even our own sun with a distance from Earth of eight light minutes (150 million kilometers) is not

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8 For text, *see* FASAN, *supra* note 2.

9 U.N.G.A. Res. 1962(18), and 1963 (18), both of Dec. 13., 1963.

10 *Principles for a Declaration with reference to the legal Nature of the Moon*, 1 PROC. COLLOQ. L OUTER SPACE 36 (1959). The first three volumes of the Proceedings on the Law of Outer Space have now been republished by the International Institute of Space Law under the auspices of its president, N. Jasentuliyana.

11 Andrew G. Haley, *Survey of Legal Opinion on Extraterrestrial Jurisdiction*, 3 PROC. COLLOQ. L. OUTER SPACE 37 (1960).

12 *Who owns the Moon?*, 88th Cong. 1st Sess. (Jan. 18, 1963).

13 For texts of the Space Treaties and Agreements, and related Documents, *see* United Nations Treaties and Principles on Outer Space, U.N. Doc. A/AC.105/572 Rev. 2 (1997).

14 The Space Treaty has up to now been signed by 120 Nations, 93 of whom have also ratified it. It is with this the practically universally accepted Legal Document regarding Outer Space and, as we have seen, the Celestial Bodies. *See* U.N.G.A., COPUOS, Report of the Legal Subcomm. (LSC), 36th Sess., Vienna, April 1-18, 1997. U.N. Doc. A/AC.105/C.2/L.206/Rev. 1 (April 4, 1997).

15 U.S. President L. B. Johnson, Letter of Transmittal of Feb. 7, 1967: *Hearings before the Senate Committee on Foreign Relations*, 90th Cong., 1st Sess. 107 (1972).

approachable for us humans. We use its radiation, without which we would not exist, but we cannot land on it, we cannot control it, and thus we can by no legal definition appropriate it: neither can a single Nation, nor the whole of humanity.

Around the sun revolve nine planets, two of which, namely Neptune and Pluto, have been discovered only recently.[16] One of them is our Earth. And at least seven of them are accompanied by one or more natural satellites, by moons, revolving around them. One of them is our own Moon.

Then there do show up occasional phenomena in the sky, which appear, grow more and more luminous, and finally disappear again: The comets. Some of them return in elliptical orbits after some decades, as comet Halley. Others seem to have nearly parabolic orbits and may after passing the sun disappear for thousands of years, or forever.[17] Here is not the place to discuss all the legends and fears connected to those objects. But one fact remains: They are Celestial Bodies, and some of them have even been investigated by terrestrial Space Ships.

Furthermore, we know that occasionally little natural objects are approaching our planet, and even fall down on it. These are the Meteorites. They are three dimensional natural objects moving in outer space. So they are bodies, most of them very small, moving within the gravitational field of our sun. Thus they (once more) are Celestial Bodies.

But that is not all. Already in 1801 Piazzi found between the orbits of Mars and Jupiter a small moving object which one year later was rediscovered by Gauss. It was obviously a very small planet in orbit around the sun, and it was called Ceres. Shortly later, Pallas was found, and then Juno and Vesta. By 1953 about 1600 such bodies, called Planetoids, or Asteroids, were cataloged, and the total number of them is estimated to be over 50,000.[18] One of them, Hermes, came nearer to the Earth's orbit than that of our own Moon. And at least one of the Asteroids, Ida, has a natural satellite of its own! A new category of Celestial Bodies was found, and was well known, even before the beginning of the Space Age.

Are then all these natural objects "Celestial Bodies" in the sense of the UNGA Resolutions, or of the Space Treaties? Is their appropriation prohibited as well as that of, say, Mars. Even if they pose threats to Earth?

There is only one hint: In the Agreement Governing the Activities of States on the Moon and other Celestial Bodies, we find a provision that *"This Agreement does not apply to extraterrestrial materials which reach the surface of the Earth by natural means."* This provision once more

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16 The astronomical facts are generally known. Reference is made to K. SCHAIFERS AND G. TRAVING, MEYERS, HANDBUCH WELTALL (6th ed., Bibliographisches Institut, Mannheim, Wien, Zurich 1984). *See also* remarks by E. Fasan in 3 PROC. COLLOQ. L. OUTER SPACE 18 (1961).

17 If their orbits are hyperbolic, they may no longer belong to our solar system.

18 *See* K. SCHAIFERS AND G. TRAVING, MEYERS, HANDBUCH WELTALL, *supra* note 16, at 162*ff*.

clarifies, *argumento e contrario,* that all other extraterrestrial material seems to be deemed "Celestial Bodies".

III. Asteroids and other small Bodies in the Literature

Already in the sixties, some Space Legal writers saw the obvious physical difference between a Moon, say of Jupiter, and an Asteroid of only a kilometer or a few hundred meters in diameter. A few opinions may be quoted in alphabetical order.

Barth calls them potential sources of raw material, especially metallics.[19] *Christol* stressed the fact that the USA position before the U.N. had made it clear that the Moon Agreement places no moratorium upon the exploitation of the natural resources on celestial bodies. But no definition of the latter term was given.[20]

Cocca wrote a most detailed investigation of the various opinions regarding the legal nature of small natural objects in Space, quoting *Csabafi, Fasan, Kiss, Markoff, Verplaetse,* and especially *Smirnoff,* Chairman of the then Working Group Three of the IISL. He endorses the solution, found by this Working Group: Celestial Bodies in the sense of this Resolution are natural objects in Outer Space, including their eventual gaseous coronas which can not be artificially moved from their natural orbits."[21]

The Committee on Commerce, Science, and Transportation of the US Senate prepared in 1980 a report on the Moon Agreement. It brought about a detailed history of the relevant UN negotiations. It started with Argentina's proposal, as presented by ambassador *Cocca,* Art. 3 of which requested a distinction of the legal system "applicable to natural resources used in their place of origin . . from those brought to Earth for use," and endorsed the "Common Heritage of Mankind" Principle.[22]

The following USSR and USA proposals are quoted both mentioning "the Moon and other Celestial Bodies" without any definition of the latter. The same must be said of the following UN draft Treaties. But that is not quite valid for the subsequent "Comparision of the Moon Agreements draft with other International Documents."[23] In discussing Art. 1, quite correctly the question is raised, "as to whether asteroids and other materials brought from deep space into earth orbit or to the earth would be subject to the term of this Agreement."[24] And under "Issues" this very same question is repeated.[25]

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19 *Das Raumzeitalter* (Dacia1981, p. 157).

20 Christol, *The Moon Treaty Enters into Force,* 79 AM.J. INT'L L. 163, at 166 (1985).

21 7 PROC. COLLOQ. L OUTER SPACE 16 (1962).

22 P. 7.

23 P. 45.

24 P. 47.

25 P. 79.

Fasan deems Asteroids not as Celestial Bodies in a legal sense, but as parts of Outer Space, as long as they can be transported into or through Space.[26] *Gorove* equals the Moon with "other planets of the solar system" which seems clearly to exclude Asteroids, Comets and Meteorites.[27] *Jenks* discussed the "Moon and other planets or satellites", the sovereignty over which should be vested with the United Nations. Radiation should be free for use. Asteroids are not mentioned specifically.[28] *Lachs* raises the question of a minimum size of a natural object in Space in order to be deemed a Celestial Body. The definition of the latter should be "land areas in Space."[29] *Haley* said in his book SPACE LAW AND GOVERNMENT: "Meteor Mining will become an industrial objective, and all the ancient problems of law will be asserted under vastly more complicated circumstances."[30] *McDougal, Lasswell and Vlasic* do point out that Asteroids might be immense deposits of useful minerals, and they quote Vassiliev who thinks that they may be used as "ocean going ships", offering "more room than . . . a man built ship."[31]

Goldie saw and clearly raised the question whether the notion of non appropriation would apply to the Planets and Moons. And then he asks: "what regimes should then govern such Planetoids as Eros, Hermes, Icarus, and others?"[32] *Hosenball* makes clear that the Moon Treaty should apply to "the list of the planets", and seems to defend these as "Celestial Bodies." He, too, does not deal with the small bodies, now under discussion.[33] *Rehm* quotes *Dettmering* and *Fasan* and their distinction between planets and Moons in orbits, unalterable by human technology.[34] And *Zhukov*, last but by no means least, strongly opposes *Fasan*, because future technology might well be able to alter the orbits of Asteroids. And he quotes *Cowle* who foresaw Asteroids, brought down to a crash on Earth, as a most dangerous weapon of the future.[35]

Finally, in "Pioneering the Space Frontier," the Report of the National Commission on Space of 1986, there is a whole chapter on "Readily accessible Asteroids," which recommended: *"Expanded Earth-based and space based searches for readily accessible asteroids; continued telescopic*

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26 FASAN, *supra* note 2, at 133.

27 STEPHEN GOROVE, SPACE LAW, ITS CHALLENGES AND PROSPECTS 174 (Sijthoff, Leiden 1977).

28 WILFRED JENKS, SPACE LAW , at 100*ff.* (Steven & Sons, London 1965).

29 Manfred H. LACHS, THE LAW OF OUTER SPACE, at 46 (Sijthoff, Leiden 1972).

30 ANDREW HALEY, SPACE LAW AND GOVERNMENT, at 133 (Appleton, Century, Crofts 1963).

31 MYRES S. MCDOUGAL, HAROLD D. LASSWELL AND IVAN A.VLASIC, LAW AND PUBLIC ORDER IN SPACE, at 767 (Yale U. Press, 1963).

32 12 PROC. COLLOQ. L. OUTER SPACE 156 (1980).

33 *See* Hosenball, *Current Issues of Space Law before the United Nations,* 2 J. SPACE L. 8 (1974).

34 HANDBUCH DES WELTRAUMRECHTS 112 (Böckstiegel ed., Carl Heymanns Verlag, Köln 1991).

35 WELTRAUMRECHT, 272*ff.* (German Translation, Berlin Verlag 1968).

characterization of their surfaces; and robotic prospector missions to particularly promising asteroids."[36]

In recent times, however, small Celestial Bodies were also seen in a new perspective. It was the perspective of such bodies coming very near to Earth, and even posing a possible threat to our planet: "*an estimated 99 asteroids or comets are known to pose possible danger to earth. While not an immediate threat, seven previously unknown asteroids are big enough and close enough to pose at least a potential threat.*"[37]

NASA started a program of "Near-Earth Asteroid Rendezvous spacecraft," which by the way discovered the Asteroid Mathilda. This so called NEAR mission will bring the spacecraft in 1999 to a distance of 18 kilometers from Eros' surface and will be able to study this "Celestial Body" in detail.[38] But also the study of comets was intensified, especially by ESA, including a factual "Lander."[39]

And we will have to be content with the following observation: "It is expected that it will take up to 30 years to discover 90% of all objects (1 km or more in size) with Earth crossing orbits."[40]

IV. Legal Definition of the Term "Celestial Body"

These astrophysical facts are of course followed by the question of definition of the Term "Celestial Body". There is no difficulty about the Moon, because our satellite is mentioned specifically in the Space Treaties. The literature quoted makes clear that the planets themselves are "Celestial Bodies" in all legal senses, too. From this follows that the main Legal provisions of the five Space Treaties and Agreements do apply to those bodies. This disregarding the fact that those five legal documents have achieved quite different adherences by Nations regarding signature, and especially, ratification. But all these instruments are in accordance concerning some basic legal rules, which may be requoted in very shortened sentences:

1) Exploration and Use of Celestial Bodies a) is to be carried out for the benefit and in the interests of all countries; b) shall be free, c) is to be conducted according to international law;

2) Celestial Bodies a) are not subject to national appropriation, b) are to be kept free from nuclear weapons or other weapons of mass destruction. c) shall be used for peaceful purposes only;

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36 *See* p. 65.

37 See Report on the 4th Space Governance Conference, held August 1997, in Denver, Colorado, referred to in 25 J. SPACE L. 172 (1997). *See also* 3 "THE NEO NEWS" of the Planetary Society (no. 3, 4th quart., 1997), which deals extensively with "Near Earth Objects".

38 HIGHLIGHTS IN SPACE 1996, at 61, 96 (U.N. ed., New York 1997). *See also* 35 AEROSPACE AMERICA 63 (Dec. 1997).

39 HIGHLIGHTS IN SPACE 1996, at 61.

40 35 AEROSPACE AMERICA at B 18 (Dec. 1997).

3) Installations (bases) a) may be erected, equipment and military personnel may be used, b) remain under jurisdiction and control of the launching state, c) are to be registered with the Secretary General of the United Nations, d) may be visited by representatives of the State Parties.

4) The notions of liability, and of mutual assistance, retention of ownership, prompt return, consultations, *etc.* are firmly written down also.

While these provisions are not contended in earnest, the question of exploitation, the notion of "Common Heritage of Mankind," of an international agency, *etc.* are still in dispute.

The above list, quoted from the Space Treaty,[41] is intentionally very brief because it does only in part touch our problem of definition.

If all these notions do apply to any land mass in Space, even, say, to a comet or asteroid with a diameter of one or ten kilometers, and if such a body is discovered to possibly cross Earths orbit, then it poses a danger not only to astronauts,[42] but also threatens to cause an adverse change of the environment of the Earth,[43] although not caused by a national activity in Space. Such a (natural) space object would be "extraterrestrial material" which would reach the surface of the earth by natural means.[44] And it would be - in all seriousness - "a phenomenon which could endanger human life."[45]

If now any state or group of states had the capability to deflect such an asteroid from its natural orbit, and guide it, for instance, towards the sun, this would of course mean intentional destruction of the said body, and nobody could or would question the legality, and the absolute necessity of such an action. But destruction of an object, a *res* in Latin legal language, is the ultimate appropriation. Surely, this could not be in earnest an act contrary to international Space Law, as it exists. So, is such a small body, for instance, a "Tunguska Meteorite," a Celestial Body in a legal sense, and would anybody be opposed to such an action on such a legal basis?

To quote quite another example: The Use of a celestial body (for peaceful purposes) is permitted. Is it permitted then, to alter the orbit of a small asteroid- as long as this does not pose any danger to earth or any of its nations? Is it permitted, to erect then on this asteroid a scientific station (of course, with due information of the Secretary General of the UN, and if necessary after consultations with other Nations). Is it then permissible to bring this Asteroid with the said station into an orbit

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41 *See supra* note 13.

42 Art. V(2) of the Space Treaty.

43 Art.1X of the Space Treaty.

44 Art. 1(1) of the Moon Agreement, *supra* note 13, which however has gained much less adherence than the other Space Treaties and Agreements (12 signatures, of those 9 ratifications only). *See supra* note 14.

45 Art. 5(3) Moon Agreement. The newly discovered Asteroid XF11 would be a typical example. An excellent overview is given by Brooks in *"Dangers from Asteroids and Comets: Relevance of International Law and the Space Treaties,"* 40 PROC. COLLOQ. L. OUTER SPACE 234*ff.* (1998)

around earth, for instance far above the geostationary orbit? It would remain free for inspection. It would remain free from all military use and from all weapons of mass destruction. But would that still be "Use" in the sense of the Space Treaties. And what, as I have once asked[46] if such an Asteroid is hollowed out, if the material thus won is used to erect stations and other structures. And if it is covered with such man made structures? If it has a diameter of a few hundred meters only, would it not somehow loose its capacity of a "natural Celestial Body", and become a man made, an artificial "space object", to which quite different legal notions would apply, for instance regarding ownership, control, registration and liability?

The makers of the Space Treaties were prudent and well advised that they dealt with "the Moon and other Celestial Bodies." But obviously they had substantial natural objects in mind, as sometimes expressly said and partly quoted above. But surely Space Law will have to proceed in this regard too. "Comet watch," "Asteroid watch," even "Meteorite watch" will not and can not remain an activity of scientific observation only. It may request, or it may at least permit positive action concerning an object thus discovered.

If we would agree with the above, we might have at least two possible means of approach. We could still call, say, Asteroids "Celestial Bodies," and wait, and try to find solutions for possible unexpected contingencies, a few examples of which are given above. Or we could in more clearness say and define what is meant by "Celestial Body" in the sense of the Space Treaties and other Instruments.

And, in due salute to the first president of the International Institute of Space Law, the late Michel Smirnoff, we could find a solution similar to that of the unanimous Draft Resolution of March 15, 1964, of the then Working Group Three of the International Institute of Space Law. The relevant paragraph might read, taking into account the legal development of the last decades, as follows:

"Celestial bodies in the sense of the treaties and agreements on outer space are natural objects in outer space including their eventual gaseous coronas which can not be artificially moved from their natural orbits."[47]

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[46] Fasan, *Large Space Structures and Celestial Bodies.* 27 PROC. COLLOQ. L. OUTER SPACE 243 (1984).

[47] Smirnoff, *Report for Working Group three of the IISL,* 7 PROC. COLLOQ. L. OUTER SPACE 352 (1964).